

Debarshi Chakraborty

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EDUCATION

- **Ramakrishna Mission Vivekananda Educational and Research Institute**
Master of Science - Big Data Analytics July 2024 - July 2026
- **Ramakrishna Mission Vivekananda Centenary College**
Bachelor of Science - Mathematics Honors September 2021 - July 2024

EXPERIENCE

- **AI/ML Intern** January 2026 - July 2026
Rezolve Ai Kolkata, West Bengal
 - Engineered a production **Agent-to-Agent (A2A)** multi-agent system end-to-end, including orchestrator design, **MCP** tool server deployment, and **Redis**-backed context sharing, powering a live retail conversational AI platform.
 - Built a **LangMem**-backed long-term memory agent injecting past user preferences into the A2A flow, cutting redundant inputs by **25%**.
 - Refactored the prompt stack into a **3-layer architecture** managed in **Langfuse**, cutting prompt-related incidents by **30%** across **6+ sub-agents**.
 - Led evaluation of multiple **LLMs** for tool calling; through careful model selection, retained routing performance while cutting latency by over **40%**.
 - Owned end-to-end implementation of autonomous **coding agents** that pull ticket details via the **Atlassian MCP** server and raise PRs automatically, streamlining the dev workflow.
- **Summer Research Intern** May 2025 - July 2025
Indian Institute of Technology, Guwahati Guwahati, India
 - Collaborated with **PhD researchers** to design **graph algorithms** for the **Popular Matching Problem** in subcubic graphs and bounded-degree graph families.
 - Developed and analyzed **combinatorial** and **reduction-based algorithms** to determine the existence of popular matchings under structural constraints; implemented prototypes in **Python**.

PROJECTS

- **Mnemos: Agentic Memory System with Provenance-Tracker Consolidation** 
 - Built a self-hosted agentic memory system with a **LangGraph-orchestrated** “sleep cycle” for extracting, deduplicating, and contradiction-checking facts from episodic history.
 - Implemented **decay-ranked retrieval**, LLM-based contradiction resolution, and **EMA confidence updates**, preserving permanent provenance chains.
 - Exposed memory operations as **MCP tools** for Claude Desktop/Cursor and built a React frontend for chat, memory inspection, and roadmap planning.
- **FinSight: Regulatory-Grade Multi-Agent Financial Intelligence System** 
 - Architected a **six-stage multi-agent RAG pipeline** (Planner, Retriever, Analyst, Auditor, Comparator, Synthesizer) via **Microsoft Semantic Kernel** with hybrid BM25 + dense retrieval and cross-encoder reranking.
 - Built an entailment-checking **Auditor agent** that hard-blocks unverifiable claims and a deterministic currency normalizer that eliminates LLM arithmetic hallucination.
 - Enforced end-to-end traceability with verbatim citations and machine-readable audit logs capturing every claim, retrieval, and agent decision.
- **Real-Time (T+0) Trade Settlement System for the US Market** 
 - Architected a 4-node permissioned blockchain network (**Hyperledger Besu, IBFT-2**) for atomic delivery-versus-payment settlement of tokenized securities.
 - Integrated **LangGraph** compliance/liquidity agents with SHAP/LIME explainability and **XGBoost** fraud detection, cutting counterparty risk by **91.7%**.
 - Reduced the settlement cycle from the industry-standard T+2 to real-time (T+0) via deterministic smart-contract execution, eliminating counterparty exposure windows.
- **Graph-Structured Deep Learning & Multi-Agent Decision Pipeline for Financial Forecasting** 
 - Proposed **THGNN-MaGNet**, a hybrid GNN (GRU-MAGE encoder, Sparse MoE, temporal-causal + global probabilistic hypergraphs) trained with a differentiable Spearman IC loss on **NIFTY 500**.
 - Achieved **rank-IC 0.0398** (+95% over baseline), **ICIR 0.420**, and **Sharpe 3.71** on a 583-day holdout backtest.
 - Built a **six-agent LangGraph pipeline** (GNN, sentiment, risk, macro, LLM report agents) with sub-45s end-to-end inference.

- **NET-LINK: Self-Healing Drone Swarms** 
 - Built a **Dec-POMDP** multi-agent environment where 5 drones maintain a dynamic communication relay under energy and collision constraints.
 - Benchmarked **MAPPO**, **MADDPG**, and **QMIX** under CTDE; MAPPO achieved **45% success** with near-zero collisions.

SKILLS SUMMARY

- **Languages:** Python, R, Java, C++, SQL
- **AI/ML:** PyTorch, Scikit-learn, NLTK, LangGraph, LangChain, OpenAI Gym, GNNs
- **MLOps & Infra:** Docker, Langfuse, Qdrant, Weaviate, FastAPI, Streamlit, Azure OpenAI
- **Data:** NumPy, Pandas, PySpark, Dask, Polars, Matplotlib
- **Tools:** Git, GitHub, VS Code, Jupyter, Neo4j

ACHIEVEMENTS

- **IEEE WCONF 2025 Publication** *2025*
Published “*Real Time Trade Settlement for the US Market*” at the 3rd World Conference on Communication & Computing. DOI: 10.1109/WCONF64849.2025.11233239
- **Cleared IIT JAM 2024** *February 2024*
Successfully cleared the IIT JAM 2024 examination in Mathematics, securing eligibility for master’s programs at IITs.

VOLUNTEER EXPERIENCE

- **Event Lead: CodeCraft — Perceptron 26, RKMVERI Annual Tech Fest** *January 2026*
Spearheaded the flagship Hackathon “CodeCraft”, managing logistics, problem statements, and participant coordination for 300+ attendees.